THE HEX UNKNOTTING NUMBER

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1. Introduction

In this paper we begin with considering a classical invariant of knot theory from the perspective of a more modern tool. The unknotting number is a classical invariant of knots defined to be the smallest number of crossing changes required to change a knot into the unknot. This invariant has been widely studied and is defined in most elementary texts on knot theory (e.g. [1], [12], or [15]). In 2008, Kauffman and Lomanaco [17] defined knot mosaics, first as a tool to study quantum knots, and later as a tool to study classical knots. The theory of knot mosaics has explored various measures of how efficiently knots can be put on mosaic diagrams ([7], [8], [9], [13], [14]). In the context of unknotting we first observe that a crossing change can be achieved through a tile rotation on a mosaic diagram of the knot. Since this operation defines the same invariant as the classical invariant, it is not particularly interesting. On the other hand, more recent work has explored mosaics composed of hexagonal tiles ([2], [3], [6], [10], [11]). Motivated by the square tile change operation, we define a new knot invariant $u_h(K)$, or the hex unknotting number, as the minimum number of hex unknotting moves given some hex mosaic diagram required to create the unknot. We explore the relationship between the hex unknotting number and the classical unknotting number through upper and lower bounds and compute values and bounds for every knot with 9 or fewer crossings. We also find a bound on the hex unknotting number based on the delta unknotting number. In addition, we calculate a large collection of examples, including the hex unknotting numbers of $(3n+1, 3)$ torus knots. Finally, we determine that the hex unknotting number is subadditive for connected sums of knots.

In the next section, we provide detailed definitions of both square and hexagonal mosaics, and the classical unknotting number. We also give a formal definition of the hex unknotting number. In section 3 we discuss the relationship between the classical unknotting number and the hex unknotting number. We provide both a lower and an upper bound for the hex unknotting number based on the knot's classical unknotting number. We also explore the relationship to the Δ-unknotting number (another unknotting invariant studied by Nakanishi [18]). This allows us to compute a range of examples.

Section 4 explores the hex unknotting of torus knots. In particular, we look at examples where the hex unknotting number is one third (or close to one third) the classical unknotting number.

A longstanding question about the classical unknotting number was whether the unknotting number was additive or subadditive under connected sum. In section 5 we address this question as it pertains to the hex unknotting number.

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Finally, in section 6, we discuss some interesting unsolved problems relating to the hex unknotting number to be considered in future research. We also include an Appendix with a tabulation of the hex unknotting number for knots up to nine crossings.

2. Background

A *knot* K is an embedding of a closed continuous loop in three space. Knots are often studied through *diagrams*. A *knot diagram* D is a two dimensional projection of the knot with over and under crossings indicated by a break in the strand. Any two diagrams of the same knot can be related by a sequence of *Reidemeister moves*, shown in Figure 1.

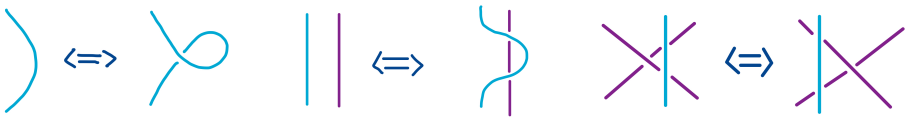


Figure 1. Reidemeister moves 1, 2, and 3

Further basics of knot theory can be found in Adams [1] or Livingston [15].

A *mosaic* is a type of knot diagram constructed with tiles. A *square n-mosaic* is a diagram constructed using square tiles, shown in Figure 2 on an $n \times n$ grid.

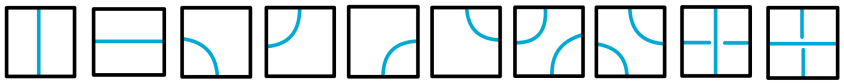


Figure 2. Tiles that can be used in a square mosaic diagram, excluding the empty tile

Knot mosaics have primarily been studied on square tiles. However, hexagonal tiles can also be used. In this research, the focus has been on hexagonal mosaics. A *hexagon mosaic* is a knot diagram constructed using the hexagonal tiles shown in Figure 3, up to rotation, on a grid.

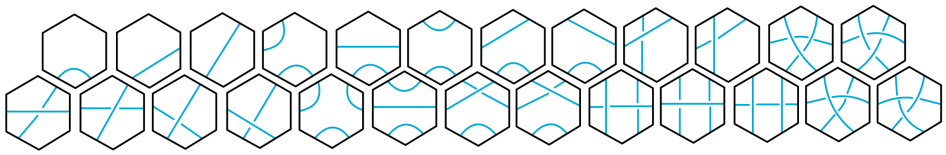


Figure 3. Tiles that can be used in a hexagon mosaic, up to rotation and excluding the empty tile. Unlike square mosaics, these include 2-crossing and 3-crossing tiles.

The *unknotting number* of a knot K is the minimum number of crossings that need to be changed on a diagram in order to produce the unknot. A crossing change allows us to swap the over- and under-strand of a specific crossing. Note that this potentially fundamentally changes our knot into another knot type, unlike

Reidemeister moves, which only change the appearance of the knot, not its knot type.

To unknot a knot we must find a set of crossing changes on a diagram that change the knot into the unknot. The crossings involved in this change are called *unknotting crossings*. For knots with unknotting numbers larger than one, there are possibly many sets of unknotting crossings. A minimal set of unknotting crossings may not exist on the crossing-minimal knot diagram, as shown by Nakanishi and Bleiler for $u(10_8) = 2$ [12].

The *square tile unknotting move*, shown in Figure 4, allows us to construct an analogous invariant to the classical unknotting number using a square mosaic diagram. The *square mosaic unknotting number*, denoted $u_s(K)$ is the minimum number of square tile unknotting moves that must be performed on some square mosaic diagram D of K to produce the unknot, where the minimum is taken over all square mosaic diagrams of K.



FIGURE 4. The square mosaic unknotting move is identical to the simple crossing change used to calculate $u(K)$. Thus $u_s(K) = u(K)$

Any knot diagram can be drawn, after planar isotopies, on a sufficiently large square mosaic tiling and crossings are isolated. Thus, $u_s(K) = u(K)$ for any knot K. This is true because, to unknot a knot on square mosaics, the tiles with the hex unknotting crossings are rotated one at a time and each tile has only one crossing. Similarly, over- and under-strands are swapped one crossing at a time for the classical unknotting number. Thus, both numbers will be the same.

This paper doesn't dedicate much time to the study of $u_s(K)$ because all of the work done in calculating $u(K)$ for knots still applies. Instead, we set our sights on a different type of mosaic.

In this paper, we approach these crossing changes on a square or hexagon grid by looking at the tiles containing crossings. Since hexagons can fit up to three crossings on a single tile, we define unknotting number of our diagram as the number of crossing tiles that need to be changed in order to produce the unknot, rather than the number of crossings that need to be changed. This allows us to cluster our crossing together to create the unknot through changes in a smaller area of the knot.

A *hexagon tile unknotting move* is a tile swap using hexagonal tiles that we can use to unknot a knot. For this, we created twelve different sets of tiles, shown in Figure 5, that can be swapped with each other as unknotting moves. Note that some of these sets use the same tiles, but different types of changes between them. Not all tile potential swaps are allowed: each of these sets of tiles follows the rule that changes must maintain *consistent connection points*. That is, endpoints of a tile that were connected in one tile in the set must also be connected in all of the other tiles in the set. This prevents the accidental creation of links when we are trying to unknot our diagrams. In order to create our sets of tiles, we followed transformations produced from reflections over lines of symmetry and rotations, usually in 120 degree intervals to maintain consistent connection points.

The *hex unknotting number* of a knot K, denoted $u_h(K)$ is the minimum number of hexagon tile unknotting moves that must be performed on some hexagon mosaic of K to produce the unknot (the minimum is taken over all hexagon mosaic of K).

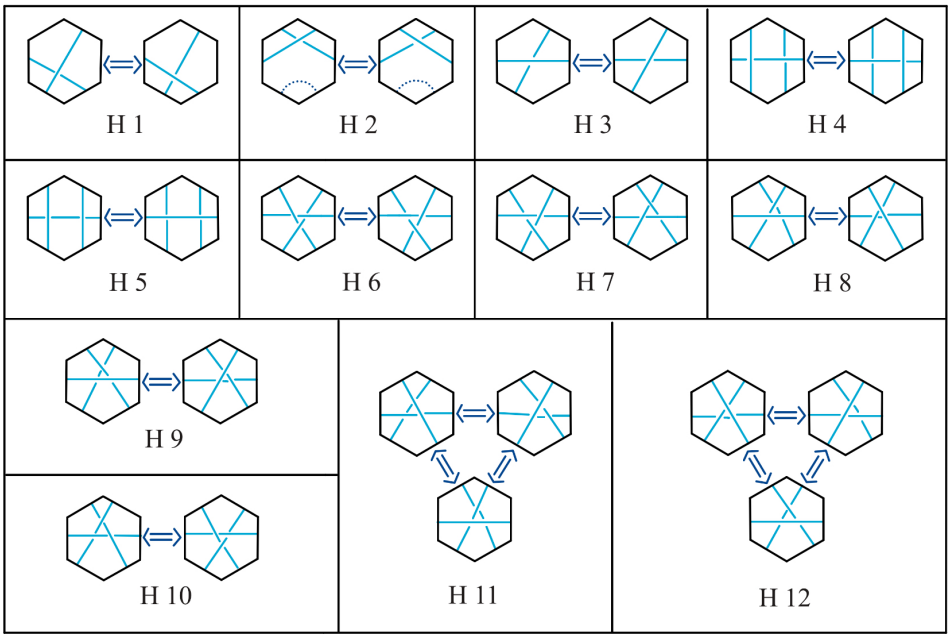


FIGURE 5. All legal tile change options when working with hexagon tiles. Unlike square mosiacs, these include 2-crossing and 3-crossing tile changes

3. Relationship to Classical Unknotting Number

Our objective is to compare hex unknotting with classical unknotting. In doing so, we also observe that the invariant $u_h(K)$ is well-defined.

Theorem 1. *For any knot, K, $u_h(K) \leq u(K)$.*

Proof. Begin with a diagram of K with a minimal set of crossing changes. By stretching the knot so that the crossings are far apart, we can draw a hex diagram with the crossings each separated on single crossing tiles. Then use moves H_1 and H_2 to perform the appropriate crossing changes. This results in a hex unknotting with the same number of tile moves as $u(K)$. □

It is clear that classical unknotting and hex unknotting produce the same result when $u(K) = 1$.

Lemma 2. *Given a knot K with unknotting number $u(K) = 1$, then $u_h(K) = u(K)$.*

Proof. When $u(K)$ is 1, only one crossing needs to be changed to achieve the unknot. As we did in Theorem 2 we can isolate the crossing on a hex tile and change it with move H_1 or H_2.

Thus, when $u(K) = 1$, then $u_h(K) = 1$. □

Things get more interesting when we can combine multiple crossing changes on a tile. Next we can determine how to construct 2-crossing tiles in which both crossings are unknotting crossings. These crossings must be consecutive, which will make unknotting more efficient. *Consecutive crossings* are crossings that occur directly next to each other along at least one strand. In a square mosaic diagram, this typically looks like two crossing tiles that are adjacent to each other. However, in a hex mosaic diagram, we can have these two crossings in the same tile.

Our objective is to take specific unknotting crossings and make them consecutive. This will allow us to put two unknotting crossings on the same hexagon tile. Fortunately, we can use Redeimeister moves to achieve this.

Lemma 3. *Any two crossings may always be placed on the same hexagon tile in some hexagon diagram.*

Proof. In a classical knot diagram, choose two crossings. Between these two crossings, some orientation exists in which the crossings are connected by a sequence of strands that doesn't connect a crossing to itself first. Between any two unknotting crossings exist a finite number of under or over-strands.

Pull the crossing strand for one crossing and towards for the other. For p strands between the two crossings, use p Reidemeister 2 and Reidemeister 3 moves to pass these other strands and make the unknotting crossings consecutive. Utilize planar isotopy as needed, keeping the pushed arc in a small neighborhood of the arc it is traveling along. Observe that we now have the two crossings consecutive. Since the pushed arc traveled close to the connecting arc, after making a crossing change we can slide the pushed arc back to its original position. Therefore, this gives an alternate diagram with the same crossing change configuration, but with the crossings consecutive.

Finally, we observe that once the crossings are consecutive, we can build a hexagon tiling with those two crossings on the same tile.

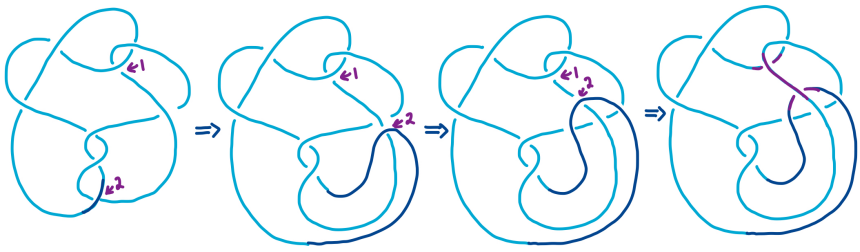


FIGURE 6. An example of moving two crossings (1 and 2) together by moving the dark blue strand, allowing the two crossings to be on the same tile when the knot K is drawn on a hexagonal grid.

□

This allows us to examine cases in which hex unknotting and square unknotting yield different results.

Theorem 4. *Given a knot, K, with unknotting number $u(K) > 1$, then $u_h(K) < u(K)$.*

Proof. From a diagram and a particular set of unknotting crossings, choose two unknotting crossings (since $u(K) > 1$, any unknotting requires at least two crossing changes). Using the algorithm described in Lemma 3, make these crossings consecutive and label this diagram D. From this design, construct a hexagon diagram D_h, which converts the two consecutive unknotting crossings into a 2-crossing tile. Then use move H_4 or H_5 to change these two crossings (and other tile moves to change any other necessary crossings).

Since both unknotting crossings occur on the same tile, the number of hexagon tile changes derived from the diagram D_h will be at least one less than $u(K)$.

Recall $u_h(K)$ is the minimum number of hexagon tile changes across all diagrams. While $u(K) - 1$ may not equal $u_h(K)$ (there may be a more efficient hex diagram), it does provide an upper bound. Thus $u_h(K) \leq u(K) - 1$, which implies $u_h(K) < u(K)$. □

This argument cannot be used to calculate $u_h(K)$ for all knots with $u(K) > 1$, but it does produce a bound for what $u_h(K)$ is at most. Three-crossing tiles allow for even more efficient unknotting, but more on that later.

These theorems follow from the properties surrounding 2-crossing unknotting tiles. This type of unknotting is analogous to the Γ-unknotting number, denoted $u_\Gamma(K)$ and defined as the number of Γ-moves to unknot some knot [4].

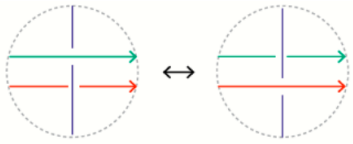


FIGURE 7. Γ-move example. This move is equivalent to H4 on a hexagon mosaic.

Three-crossing tiles have much more variety than 2-crossing tiles. They allow for some knots to achieve the lowest possible bound for the hexagon unknotting number. Since a 3-crossing tile changes at most 3 crossings at once, we have a lower bound for the efficiency of hex unknotting.

Theorem 5. *Given a knot K with unknotting number $u_s(K)$, then $u_h(K) \geq \frac{1}{3}u(K)$*

Proof. When drawing a knot on a hexagonal mosaic diagram, there can be a maximum of 3 crossings on a single tile. This means that one hexagon unknotting move will change a maximum of 3 crossings. If we implement the crossing changes ignoring the existence of tiles we still accomplish an unknotting, so we have $u(K) \leq 3u_h(K)$. So, the greatest improvement that we can make to our unknotting number using hexagons is $\frac{1}{3}$ the classical unknotting number.

□

Corollary 6. *The hexagon unknotting number is bounded by $\frac{1}{3}u(K) \leq u_h(K) \leq u(K)$.*

Proof. Theorem 1 defines the upper bound. Theorem 5 defines $\frac{1}{3}u(K) \leq u_h(K)$ as a lower bound. □

In some instances, a hex mosaic diagram can immediately be constructed with a 3-crossing unknotting tile. An example is taken from this square diagram of 8_{19}, in which three unknotting crossings form the necessary triangular configuration for a 3-crossing hexagon tile in Figure 8. It is clear that $u_h(8_{19}) = 1$.

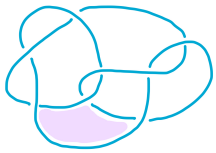
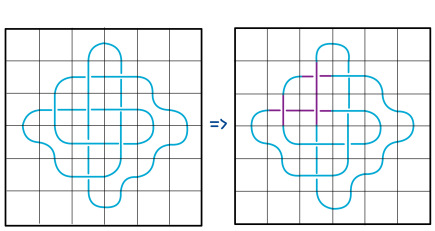
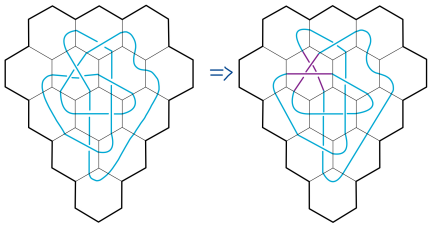


FIGURE 8. 8_{19} with highlighted triangular configuration



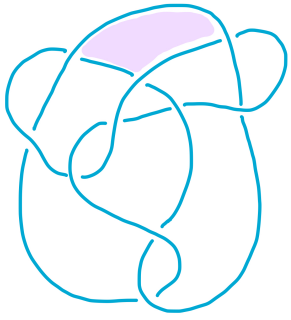
(A) 8_{19} has a set of three unknotting crossings forming a triangular configuration that produces the unknot when all crossings are changed.



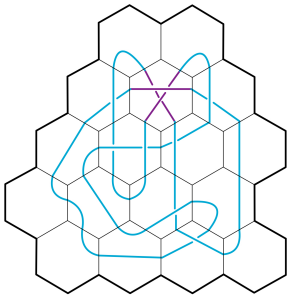
(B) 8_{19} hexagonal mosaic derived from the square mosaic that converts a triangular configuration into a 3-crossing tile using H8. This calculates $u_h(8_{19}) = 1$

FIGURE 9. 8_{19} on square and hexagonal grids.

In some instances, such as 10_{161}, a sequence of Reideimester moves and planar isotopies reveals the diagram in which this triangle configuration is clear. Figure 10b offers visual proof. Conversion into a 3-crossing unknotting hexagon tile is easy, and we observe $u_h(10_{161}) = 1$.



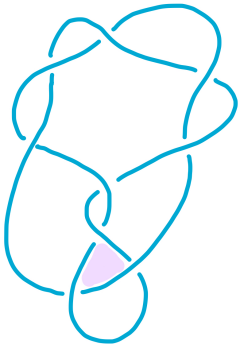
(A) 10_{161} diagram following a Redeimester 3 move.



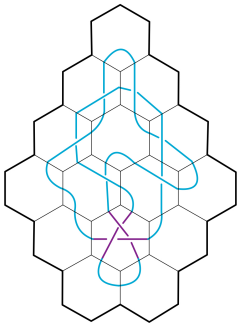
(B) 10_{161} hexagonal mosaic uses H8 to calculate $u_h(10_{161}) = 1$.

There are also some knots where there is a clear triangle of crossings, but these crossings don't allow us to unknot the knot in a single move. An example of this is the 9_6 knot, which has an unknotting number of 3 and a clear triangle of crossings

within the diagram. Unfortunately, the crossings that form this triangle aren't a set of unknotting crossings. In this diagram, the unknotting crossings need to be spread around the diagram more, making it difficult to put the unknotting crossings together in a small space.



(A) 9_6 minimal diagram with only triangular configuration highlighted



(B) 9_6 hexagonal mosaic. Unknotting the 3-crossing tile using H6 doesn't produce the unknot, so this may not be the unknotting diagram.

Some knots don't have a clear diagram nor an obvious sequence of Redeimester moves to solidify this 3-crossing unknotting tile. A good example of this is 9_3, which has no immediate 3-crossing configurations and a challenging or potentially impossible sequence of Redeimester moves to construct a hexagon unknotting diagram.

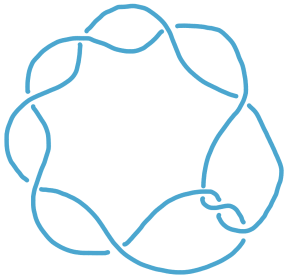


FIGURE 12. 9_3 diagram

To evaluate 9_3 and other similar knots that don't have a clear hex unknotting number, we compare the unknotting move that is analogous to the 3-crossing unknotting tile, which is the Δ-move and the Δ-unknotting number $u_\Delta(K)$, which is the minimum number of Δ-moves to convert some knot K into the unknot [4].

The Δ-move is studied in depth and $u_\Delta(K)$ is calculated for many knots in [18]. With $L_x \to L_y$ being labeled as the Δ_{xy} move, they used the basis of L_1, L_2, L_3, and L_4 and described each move as a sequence of other moves [18]. By adding L_a, L_b, L_c, and L_d to account for every possible 3-crossing up to symmetry, we find the

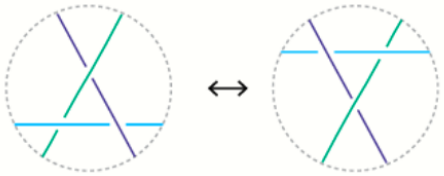


FIGURE 13. Δ-move example using H which can also be represented by H_{10}. For clarity in 16, we reference these by their Δ-move naming scheme as opposed to their H move naming scheme.

basis $x, y \in \{a, b, c, d, 1, 2, 3, 4\}$. We then relate every possible 3-crossing hexagon unknotting move to every possible Δ-move. Δ_{14} and Δ_{23} are Δ-moves, while hex unknotting moves consist of $\Delta_{ad}, \Delta_{bc}, \Delta_{13}$ and Δ_{24}. We know from Nakanishi that

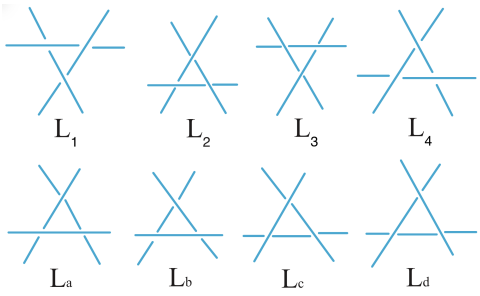


FIGURE 14. All possible 3-crossing links. We use these labels to define Δ labels. These are directly related to a valid hexagon move or indirectly related to any hexagon move using a sequence of those moves.

any two Δ_{xy} moves are related by a finite sequence of Δ-moves and Redeimester moves [18]. We constructed the additional explanation for $\Delta_{ad} \cong H_8$ and $\Delta_{bc} \cong H_9$.

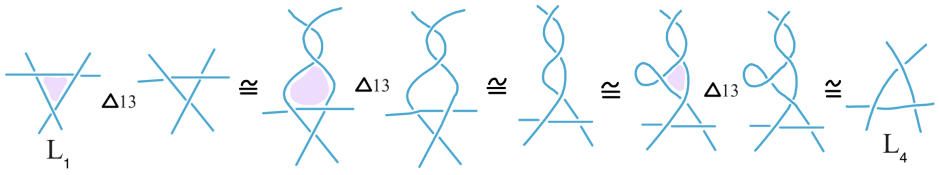


FIGURE 15. Δ_{14} as a sequence of Δ_{13}. We haven't found a method to show Δ_{13} as a sequence of Δ_{14}, but we confirm $\Delta_{14} \to 3\Delta_{12}$ and $\Delta_{12} \to 2H_6$, implying $\Delta_{14} \to 6H_6$, showing these moves are related sequentially by six moves of each other.

By counting the number of moves between each sequence, we relate the Δ-moves, which are Δ_{14} and Δ_{23}, to the hexagon unknotting moves, which are $\Delta_{ad}, \Delta_{bc}, \Delta_{13}$, and Δ_{24}.

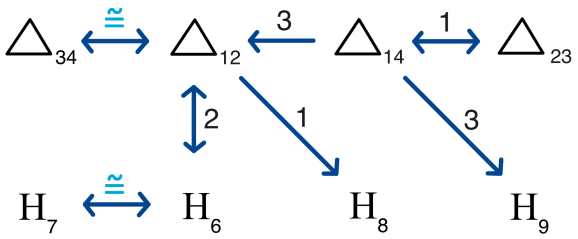


FIGURE 16. Δ_{14} and Δ_{23} are valid Δ-moves, while Δ_{12} and Δ_{34} are invalid Δ-moves. We generate these relations by computing them ourselves with the sequencing method shown in Figure 15 or using prior work from [18]. Note that direction confirms two unknotting moves are in the same equivalence class, but the number of moves in the finite sequence isn't necessarily symmetric [18].

The move count multiplies when taken across a Δ-move relation, since defining any move requires us to construct all other moves as a sequence of the chosen move. We conclude any H-move is described by a sequence of 6 or fewer Δ-moves, and vice versa.

Theorem 7. *Given some knot K with $u(K) = 3$, if $u_\Delta(K) > 6$, then $u_h(K) > 1$.*

Proof. Suppose the knot with $u_h(K) = 1$. This implies one hexagon move successfully unknots K. This move can be represented with some sequence of no more than 6 delta moves, so $u_\Delta(K) \leq 6$. However, $u_\Delta(K) > 6$, showing a contradiction. □

With a theorem to assist us, we examine 9_3 and other knots with $u_s(K) = 3$. Since $u(9_3) = 3$ and $u_\Delta(9_3) = 9$, we may conclude $u_h(9_3) > 1$. By our earlier bounding formula, we know $u_h(9_3) < u(9_3) = 3$, implying $u_h(9_3) = 2$.

This result implies that no diagram exists in which a 3-crossing unknotting tile can be designed. Many knots behave this way, confirming our suspicions. This shows the vast difference in behavior for 2-crossing and 3-crossing unknotting. Two-crossing tiles will always be achievable, but 3-crossing tiles have significant limitations as defined by the Δ-unknotting number.

We apply this same rationale to a number of knots with under 10 crossings and categorize almost all of them. The results are summarized in the Appendix. Two main outliers are 7_1 and 9_{16}. Both have square unknotting numbers of 3 and Δ-unknotting numbers of 6, so Theorem 8 doesn't apply. While we have a lower bound using Theorem 5, we still don't believe 7_1 has a 3-crossing unknotting tile, but we need a different method (or a sharper bound on the sequential Δ-move to H-move relation) to determine this unknotting number.

4. Torus Knots

Theorem 8. *For torus knots $T_{3n+1,3}$, $u_h(T_{3n+1,3}) = n$.*

Proof. It follows from the torus knot unknotting number formula that $u(T_{3n+1,3}) = 3n$, or $u_s(T_{3n+1,3}) = 3n$. Using the minimal crossing diagram of $T_{3n+1,3}$, find n distinct triple intersections most close to the interior, and construct hex diagrams such that each triple intersection is contained in a singular tile. Unknot accordingly. □

When comparing torus knots, it seems that $T_{3n+1,3}$ torus knots maximize the efficiency of the hexagon diagram with the 3-crossing unknotting tiles.



(A) $T_{4,3}$ (B) $T_{7,3}$ (C) $T_{10,3}$

FIGURE 17. $T_{3n+1,3}$ knots generated using Wolfram Mathematica

5. SUBADDITIVITY

Theorem 9. *For a knot K with connected sum $K = K_1 \# K_2$, $u_h(K) \leq u_h(K_1) + u_h(K_2)$*

Proof. Suppose some set of unknotting crossings for K. For $K = K_1 \# K_2$, map the unknotting crossings of K to the respective crossings in K_1 or K_2. Since an additional crossing change in K_1 or K_2 may be necessary, we produce the following inequality. □

We can't discuss the unknotting number without analyzing unknotting across connected sums. For a connected sum $K = K_1 \# K_2$, it is well known among knot theorists that the unknotting number is subadditive across connected sum, or $u(K) \leq u(K_1) + u(K_2)$. It was conjectured that $u(K) = u(K_1) + u(K_2)$, but this was recently disproven [5]. We have a distinct disproof for this conjecture on the hexagon unknotting number. Take the connected sum of 9_3 and 3_1, yielding the following inequality:

$$u_h(9_3 \# 3_1) = 2 < 3 = 2 + 1 = u_h(9_3) + u_h(3_1)$$

Since some knots allow for 3-crossing unknotting tiles while others don't, we conclude the hexagon unknotting number must be strictly subadditive.

Conjecture 10. *For $T(2, 2n+1)$ torus knots, $u_h(K) = \lceil \frac{u_s(K)}{2} \rceil$.*

With the $T(2, 2n+1)$ torus knots, the classical unknotting number is calculated by taking the number of crossings minus one and dividing by 2. We know that we can always put two crossings together on one tile, which lets us cut our crossing number in half, rounded up. We haven't found any examples where this bound can be improved or changed because of the structure of $T(2, 2n+1)$ torus knots, but we haven't yet found a proof that this is true, so it is still a conjecture right now.

6. Future Work

Our research primarily examined low crossing number knots, meaning most of our knots had unknotting numbers of 3 or fewer. The only knot in our appendix with a larger square mosaic unknotting number than 3 is 9_1. So our unanswered question is on the properties of higher unknotting number knots.

Consider any knot with unknotting number 4. In this instance, we find a hexagon unknotting number of 2 using two 2-crossing unknotting tiles or one 3-crossing unknotting tile and one 1-crossing unknotting tile, if possible.

Now consider a knot with unknotting number 5 or more. We find a hexagon unknotting number of 2 with a 2-crossing and 3-crossing unknotting tile or two 3-crossing unknotting tiles. Both depend on the Δ-unknotting number. For a knot with $u_s(K) \in [4,6]$, if $u_\Delta(K) > 12$, then $u_h(K) > 2$. We assert $u_s(K) \in [4,6] \Rightarrow u_h(K) \in [2,3]$ by generalizing our process from Lemma 5, or using the Γ-unknotting number. We also generalize the Δ-unknotting number bounding process for any $u_s(K)$ with a 3 divisor. This allows us to construct some bounding with these unknotting numbers.

Our goal is to describe a functional relationship from $u_h(K)$ to $u_\Gamma(K)$ and $u_\Delta(K)$. Assuming $u_\Gamma(K)$ halves the unknotting number and $u_\Delta(K)$ is bounded by five times the hexagon unknotting number - since six is the cusp - we suggest an upper bound.

Conjecture 11. $u_h(K) \leq \min\{u_\Gamma(K), \lceil \frac{u_\Delta(K)}{5} \rceil\}$

However, a more nuanced bound or even a formula could be constructed to identify the relationship between these unknotting numbers.

7. Appendix

Below is a table of known information about the hex unknotting number for knots up to nine crossings. We also include for reference the classical unknotting number and the delta unknotting number, as indicated by KnotInfo [16]. Many of these calculations are a direct result of Lemma 2 or Lemma 3. Lemma 3 also allows us to determine that the hex unknotting number of 9_1 is 2 since the classical unknotting number is 4.

In the case of knots with unknotting number three, some can be obstructed from having hex unknotting number 1 by Theorem 7. This includes 9_3, 9_6, 9_9, 9_{10}, 9_{13}, 9_{35}. The knot 8_{19} has hex unknotting number 1, as shown in Figure 9b. Below we include diagrams for the hex unknotting number one knots 9_{38} and 9_{49}.

The hex unknotting numbers of the knots 7_1 and 9_{16} could not be determined with the tools we have developed so far. We provide a range of possible values.

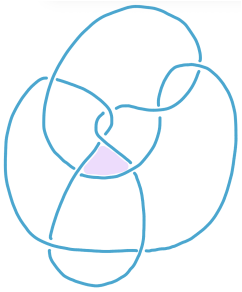


FIGURE 18. 9_{38} knot with hex unknotting crossings highlighted.



FIGURE 19. 9_{49} knot with hex unknotting crossings highlighted.

K	$u(K)$	$u_\Delta(K)$	$u_h(K)$	K	$u(K)$	$u_\Delta(K)$	$u_h(K)$
3_1	1	1	1	9_8	2	2	1
4_1	1	1	1	9_9	3	8	2
5_1	2	3	1	9_{10}	3	8	2
5_2	1	2	1	9_{11}	2	4	1
6_1	1	2	1	9_{12}	1	1	1
6_2	1	1	1	9_{13}	3	7	2
6_3	1	1	1	9_{14}	1	1	1
7_1	3	6	[1,2]	9_{15}	2	2	1
7_2	1	3	1	9_{16}	3	6	[1,2]
7_3	2	5	1	9_{17}	2	2	1
7_4	2	4	1	9_{18}	2	6	1
7_5	2	4	1	9_{19}	1	2	1
7_6	1	1	1	9_{20}	2	2	1
7_7	1	1	1	9_{21}	1	3	1
8_1	1	3	1	9_{22}	1	1	1
8_2	2	2	1	9_{23}	2	5	1
8_3	2	4	1	9_{24}	1	1	1
8_4	2	3	1	9_{25}	2	2	1
8_5	2	3	1	9_{26}	1	2	1
8_6	2	2	1	9_{27}	1	2	1
8_7	1	2	1	9_{28}	1	1	1
8_8	2	2	1	9_{29}	2	1	1
8_9	1	2	1	9_{30}	1	1	1
8_{10}	2	3	1	9_{31}	2	2	1
8_{11}	1	1	1	9_{32}	2	1	1
8_{12}	2	3	1	9_{33}	1	1	1
8_{13}	1	1	1	9_{34}	1	1	1
8_{14}	1	2	1	9_{35}	3	7	2
8_{15}	2	4	1	9_{36}	2	3	1
8_{16}	2	1	1	9_{37}	2	3	1
8_{17}	1	1	1	9_{38}	3	6	1
8_{18}	2	1	1	9_{39}	1	2	1
8_{19}	3	5	1	9_{40}	2	1	1

References

[1] Adams, C. *The Knot Book*. American Mathematical Society, Providence, Rhode Island, 2004.

[2] Alanis, C., Allen, R., Gilbert, M., Recker, M., and Kearney, K. Hex mosaic diagrams are more tile efficient than square mosaic diagrams. *The PUMP Journal of Undergraduate Research 9*, null (2026), 17–29.

[3] Alexander, M., Foster, S., Krakoff, G., and McLoud-Mann, J. Hexagonal mosaic knots, 2017.

[4] Ali, D., Yang, Z., and Sheikh, M. I. A new unknotting operation for classical and welded links. *arXiv preprint arXiv:2208.08761* (2022).

[5] Brittenham, M., and Hermiller, S. Unknotting number is not additive under connected sum. *arXiv preprint arXiv:2506.24088* (2025).

[6] Bush, J., Commins, P., Gomez, T., and McLoud-Mann, J. Hexagonal mosaic links generated by saturation. *Journal of Knot Theory and its Ramifications 29*, no. 12 (2020), 2050085–2050085.

[7] Heap, A., Baldwin, D., Canning, J., and Vinal, G. Tabulating knot mosaics: Crossing number 10 or less, 2023.

[8] Heap, A., and Knowles, D. Tile number and space-efficient knot mosaics. *Journal of Knot Theory and Its Ramifications 27*, 06 (2018), 1850041.

[9] Heap, A., and Knowles, D. Space-efficient knot mosaics for prime knots with mosaic number 6. *Involve 12*, 5 (2019), 767–789.

[10] Howards, H., Li, J., and Liu, X. An infinite family of knots whose hexagonal mosaic number is only realized in non-reduced projections, 2019.

[11] Howards, H., Li, J., and Liu, X. Bounding crossing numbers in hexagonal mosaics. *Kobe Journal of Mathematics 41*, null (2024), 41–72.

[12] Johnson, I., and Henrich, A. K. *An interactive introduction to knot theory*. Courier Dover Publications, 2017.

[13] Lee, H. J., Hong, K., Lee, H., and Oh, S. Mosaic number of knots. *Journal of Knot Theory and its Ramifications 23*, 13 (2014), 1450069.

[14] Lee, H. J., Ludwig, L., Paat, J., and Peiffer, A. Knot mosaic tabulation. *Involve. A Journal of Mathematics 11*, 1 (2018), 13.

[15] Livingston, C. *Knot Theory*, vol. 24 of *Carus Mathematical Monographs*. Mathematical Association of America, Washington, DC, 1993.

[16] Livingston, C., and Moore, A. H. Knotinfo: Table of knot invariants. URL: `knotinfo.org`, Current Month Current Year.

[17] Lomonaco, S. J., and Kauffman, L. H. Quantum knots and mosaics. *Quantum Information Processing 7*, 2-3 (2008), 85.

[18] Nakanishi, Y. Replacements in the conway third identity. *Tokyo journal of mathematics 14*, 1 (1991), 197–203.